

# All C++ Operators — Written by Qasim Niamat

This comprehensive guide provides step-by-step explanations and examples for all C++ operators, organized by category for clarity and learning efficiency.

## 1. Arithmetic Operators

+ (Addition): Adds two values. Example: `int sum = a + b;`  
- (Subtraction): Subtracts right operand from left. Example: `int diff = a - b;`  
\* (Multiplication): Multiplies two numbers. Example: `int prod = a * b;`  
/ (Division): Divides left operand by right. Example: `double q = 7.0 / 2.0;`  
% (Modulo): Gives remainder. Example: `int r = 7 % 2;`

## 2. Unary Operators

++ / -- : Increment or decrement values (prefix or postfix).  
! : Logical NOT.  
~ : Bitwise NOT.  
& : Address-of operator.  
\* : Dereference operator.  
sizeof : Returns the size of a data type.

## 3. Assignment Operators

= : Assign value. Example: `a = 5;`  
+=, -=, \*=, /=, %= : Compound assignment operators.

## 4. Relational Operators

== : Equal to.  
!= : Not equal to.  
<, <=, >, >= : Comparison operators returning boolean values.

## 5. Logical Operators

&& : Logical AND.  
|| : Logical OR.  
! : Logical NOT.

## 6. Bitwise Operators

& : Bitwise AND.  
| : Bitwise OR.  
^ : Bitwise XOR.  
~ : Bitwise NOT.  
<< : Left shift.  
>> : Right shift.

## 7. Pointer and Member Operators

. : Member access through object.

-> : Member access through pointer.

& : Address-of.

\* : Dereference.

->\* / .\* : Access member via pointer-to-member.

## 8. Scope Resolution Operator

:: : Used to define scope (namespace, class, or global).

## 9. Conditional (Ternary) Operator

? : condition ? expr1 : expr2 — returns one of two values based on condition.

## 10. Comma Operator

, : Evaluates expressions left to right and returns the last one.

## 11. Memory Operators

new / delete : Allocate and deallocate single objects.

new[] / delete[] : Allocate and deallocate arrays.

## 12. Type Cast Operators

static\_cast : Compile-time safe conversion.

dynamic\_cast : Runtime-checked cast for polymorphism.

reinterpret\_cast : Bit-level cast (use carefully).

const\_cast : Add or remove const/volatile qualifiers.

## 13. typeid Operator

typeid : Returns runtime type information of an expression.

## 14. Operator Overloading

Allows defining custom meanings for operators for user-defined types.

## Conclusion

This document, prepared by Qasim Niamat, encapsulates all core C++ operators with real-world examples and concise descriptions. It serves as an essential learning resource for both beginners and intermediate-level C++ programmers.